Spam email detection using BERT and GPT models

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Abstract

Spam email detection remains a critical challenge in cybersecurity, as malicious emails continue to evolve in sophistication. Traditional methods often rely on handcrafted features or shallow machine learning models, which struggle to capture the complex semantic and contextual patterns in modern spam emails. In this paper, we try to implement a spam email detection system using the GPT model while using BERT as the base model to find out the potential performance improvement of the GPT model.

After comparing the performance of the two models, we found that the BERT model can greatly improve the performance of the spam detection rate, and for GPT model, training time increases a lot and result is not good compared to the BERT model.

1 Introduction

With the rapid development of the Internet, email has become a fast and convenient method of communication that is widely used in our daily lives.

However, the popularity of email has also caused a problem that cannot be ignored: spam. Unscrupulous businessmen and criminals have exploited this communication medium to send spam with commercial advertisements and malicious or illegal files to users on a large scale (Kumar et al., 2020). Those spam emails not only occupy network resources and cause network congestion, but also affect people's normal work and life.

Therefore, how to quickly and accurately check whether an email is spam or not becomes a major topic worthy of research. It can protect people's property and privacy security and also improve people's work and life efficiency.

In this paper, we will try to solve the spam email detection problem using the latest transformer based models, and using the TREC 2007 spam email dataset to train our model (Cormack, 2007). Considering the growing use of pre-trained language in the NLP systems, in this experiment, we will adopt two distinct transformer based models, BERT(encoder-only model) (Devlin et al., 2018) and GPT-2(decoder-only model) (Radford et al., 2019), for this spam detection task, and to verify their real performance on our task. This comparative study will also try to find out if the generative pretraining (GPT-2) model can provide any advantages or disadvantages in a spam detection problem.

2 Theory

2.1 Transformer & Models Used

The transformer architecture used in the natural language processing (NLP) field is a revolutionary technology. By replacing traditional recurrent and convolutional layers with a mechanism based on self-attention (Vaswani et al., 2023), it enables the model can capture the contextual relationships more effectively than previous architectures.

2.2 BERT & GPT-2

The models we will use in this study are two popular transformer-based models, BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer), they are two distinct models with different component stacks, and they are designed with different approaches. BERT (Koroteev, 2021) is an encoder-only model based on a bidirectional deep learning model and can capture the information from both directions, making it suitable to understand language structure and can handle tasks like question answering, named entity recognition etc. In contrast, GPT and its new versions (GPT-2, GPT-3, and GPT-4) are decoder-only models; they focus on an auto-regressive approach (Zhao et al., 2025), and its token is based on a left-to-right manner, which makes it more suitable to generate creative content.

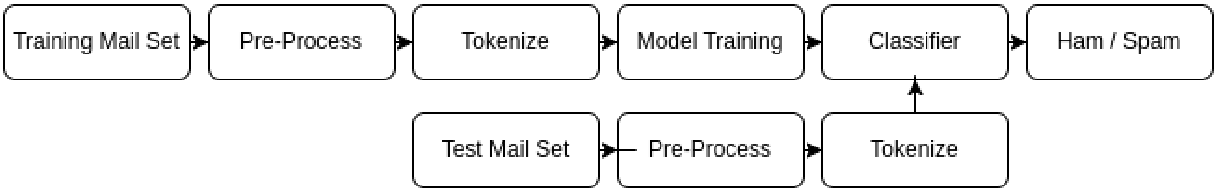

Figure 1: Spam email classification process

2.3 Evaluation Metrics

We will use the commonly used accuracy, precision, recall, and F1-Score as evaluation indicators on the test set, which can provide clear and interpretable metrics for assessing classification performance, we will also compare train loss and test loss to find the most suitable epoch number.

3 Data

This paper uses the trec07 spam corpus(Cormack, 2007) as the experimental dataset downloaded from Kaggle, and it is a public dataset provided by the International Conference on Text Retrieval. The trec07 dataset includes 29923 spam emails(56%) and 23745 ham emails(44%). This paper randomly chooses 80% as training data and uses the remaining 20% as test data.

The data columns are as follows:

- **label**: Email label, 1 for spam / 0 for ham.
- **origin**: Email content, containing all the text in an email.

4 Method

4.1 Data Preprocessing

The dataset does not suffer from significant class imbalance (56% vs 44%), so extra dataset balance operation will not be executed, and the data pre processing steps used in this paper will only including the following steps:

- Replace the abbreviations with their normal form.
- Remove all the stop words.
- Lemmatize the words with it's normal form.

We will use functions provided by the NLTK library to remove the stop words and lemmatize the words. Meanwhile, we also use the regular expressions library to replace the abbreviations with their normal form to make the data more clean and easy to process. Then the data will be tokenized and converted into a format that can be used by the model. The processed data will be split into training(80%) and testing data(20%) accordingly.

4.2 Email spam text classification process

The Email spam text classification process is shown in Figure [1] below. The process includes the following steps:

- Preprocessing the data
- Tokenization
- Model building and training
- Spam email classification

Loss value will be calculated every epoch for both training and test datasets, but accuracy, precision, recall, and F1-Score will available only on test dataset. With this in mind, we will use 20 epoches for each model.

4.3 BERT model

The BERT model used here is a pre-trained model sentence-transformers/bert-base-nli-mean-tokens(Reimers and Gurevych, 2019), which can be downloaded from the huggingface model hub. It contains 2 linear layers with 256 hidden units, dropout rate set to 0.2,and use RELU function as activation function. Loss function used in BERT model is Binary Cross Entropy Loss, and optimizer used is Adam optimizer with default learning rate.

4.4 GPT model

GPT model, on the other hand, use the pre-trained model gpt2 provided OpenAI(Radford et al., 2019). However, GPT model has different input with BERT model, so input data will modified to fit the GPT model accordingly. Loss function used in GPT model is Cross Entropy Loss instead of Binary Cross Entropy Loss, and optimizer used is Adam optimizer with default learning rate.GPT

Figure 2: BERT Model Loss And Accuracy

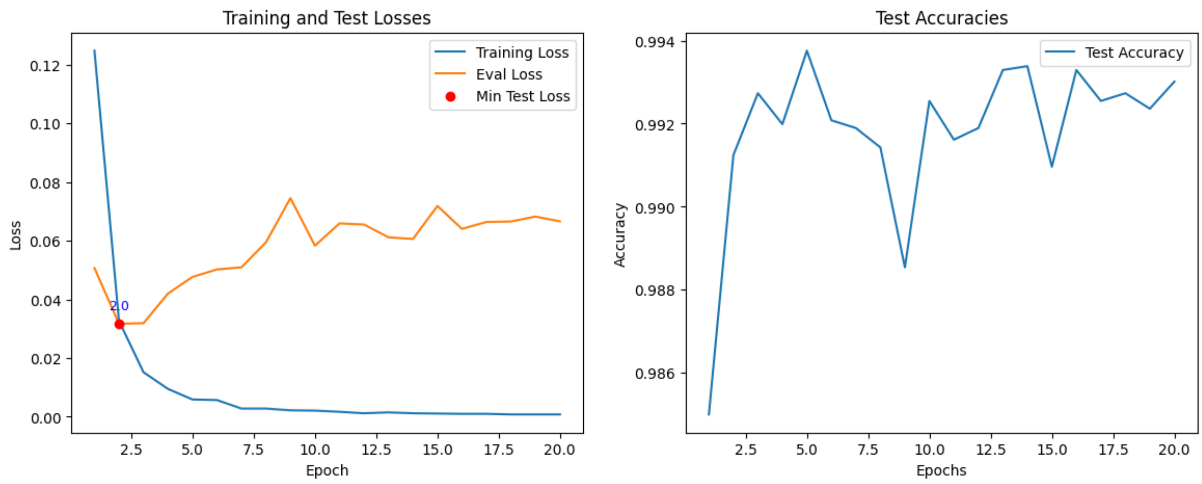

Figure 3: BERT Model Loss And Accuracy

model contains 2 linear layers with 256 hidden units, dropout rate set to 0.2, and use RELU function as activation function.

4.5 Evaluation

This article uses following four values as evaluation indicators. 1.Accuracy. 2.Precision. 3.Recall. 4.F1-score. Accuracy represents the spam detection rate, which is the accuracy of the model's classification results, Precision represents the accuracy of the classifier classification, recall represents the spam detection rate and F1 value is a combination of the precision and recall of the classifier, which is the harmonic average of the two, and it is a value between 0 and 1.

5 Results

5.1 Experiment Environment

The experiments in this paper are conducted on Google Colab(Google) with an NVIDIA GeForce A100 40G graphics card to speed up the training time. The main libraries used in this experiment includes nltk, pytorch, scikit-learn, transformers and matplotlib libraries. For the code related to this experiment, please refer to the following link.[1]

5.2 Experiment Results

In this experiment, we employed the BERT model, its loss and accuracy are illustrated in Figure [2]. The plotted results indicate that the training loss consistently decreased from 0.14 to around 0.02. Meanwhile, the test loss shows a decreasing trend at the beginning from around 0.11 to around 0.07.

[1] Source Code:https://github.com/qqyfly/text_mining_project

Model	BERT	GPT-2
Epoch	11	2
Accuracy	0.9761	0.991243
Precision	0.9758	0.991818
Recall	0.9814	0.992481
F1-score	0.9786	0.992150

Table 1: BERT And GPT Model Evaluation

However, after 11 epochs, an increase in the test loss was observed, suggesting potential overfitting.

For the accuracy on test data, test accuracy increased from around 0.965 to 0.978, with some fluctuations on epoch 15. From the plot, we can say that the BERT model archive a pretty good performance.

For the GPT-2 model, in Figure [3], we notice that the training loss decreases sharply from 0.12 to around 0.01 in 3 epochs, and then decreases slowly and approach 0 until epoch 20. However, we also notice that test loss decreases at the beginning, but after epoch 2, it begins to increase rapidly with some fluctuations. It is potential overfitting.

We list the final four values of the BERT and GPT2 models in Table [1] using the corresponding value of epoch number that minimize the test loss.

6 Discussion

From the previous result, we notice that BERT model perform much better than the GPT model in the spam email detection task. This is because of the following reasons.

1. Fundamental architectural differences between the two models.

We know that BERT Uses Bidirectional Context, meaning it considers both the left and right context when encoding a word. This allows BERT to capture rich contextual information about the entire email, making it well-suited for classification tasks where understanding the whole text is important.

GPT-2 is an autoregressive model, meaning it generates text from left to right, only considering past tokens. This limits its ability to understand the full context of an email, especially when key spam indicators appear later in the text.

So from the result, we can say that BERT is more suitable to handle classification tasks than GPT-2.

2. Effective Epochs and Training Time

In our experience, even with 20 epochs, BERT can run much faster, in around 10 minutes, while for GPT-2 model, the whole processing time will around 30 minutes, even if we set the per_device_train_batch_size and gradient_accumulation_steps to a relative large number, which is 64 and 2, respectively.

In the GPT-2 experiment, we also found that after 2 epoches, the test loss began to increase, which make sense, because in most of the LLM supervised training, we will only train the input 2-4 round, not 20 in our case, which lead to the overfiting issue in our GPT-2 experiment.

Regarding the limitations of this work, we can point out the following:

1. Limited Dataset

We only use the text data instead of html mail which may contains more multimedia information, such as picture and video, so the model we use may not be able to detect spam emails with multimedia information.

2. Limited Models

We only use the BERT and GPT model to detect spam emails, and there are many other models that can be used to detect spam emails, like DistilBERT (Sanh et al., 2019), RoBERTa (Liu et al., 2019), ALBERT (Lan et al., 2019), which may generate better performance than BERT.

About the future work and the limitations mentioned above, we need use more advanced models to detect spam emails, to identify those spam emails with multimedia information.

7 Conclusion

In this study, we aimed to compare the performance of BERT and GPT-2 in the task of spam email classification.

7.1 Key Findings

After training and evaluating both models, we found that BERT significantly outperformed GPT-2 in classifying spam emails. The primary reasons for this performance gap from the fundamental differences in how these models process text:

- Bidirectional vs. Unidirectional Processing:

  BERT's check both past and future tokens in a text so it can get a richer contextual information. In contrast, GPT-2's left-to-right processing limited its ability to capture later token information, which let the model know less context information about the email.

- Optimization for Classification:

BERT's pretraining objectives made it suitable for classification tasks. GPT-2, primarily designed for text generation, making it less effective for classification.

- Training Time Variability Among Transformer-Based Models :

  Even if those two models are Transformer-Based, the training time will be different, and we should choose a suitable model for a specific task instead of using a generalised model.

7.2 Extent to Which the Problem Was Solved

In addition to comparing the performance of BERT and GPT-2, we also gained several important insights from this study:

- Model selection is crucial:

  Choosing the right Transformer model for a specific natural language processing (NLP) task is crucial. GPT-2 performs well in text generation, while BERT is more efficient in classification tasks.

- Pre-training objectives affect model performance:

  The pre-training method of the model can significantly affect the performance of its downstream tasks. Classification tasks often benefit from bidirectional training objectives.

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